MATHÉMATIQUES
VISION
APPRENTISSAGE

– LLM –
Hierarchical language models

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27/03/2025

INTRODUCTION

Goals :

- Improve the understanding of (very) long-term context
- Reason at different levels of abstraction.

Proposed solutions:

- XLNet (Yang et al.)
- Hierarchical Transformers (Pappagari et al.)
- Hierarchical Embeddings (ours)

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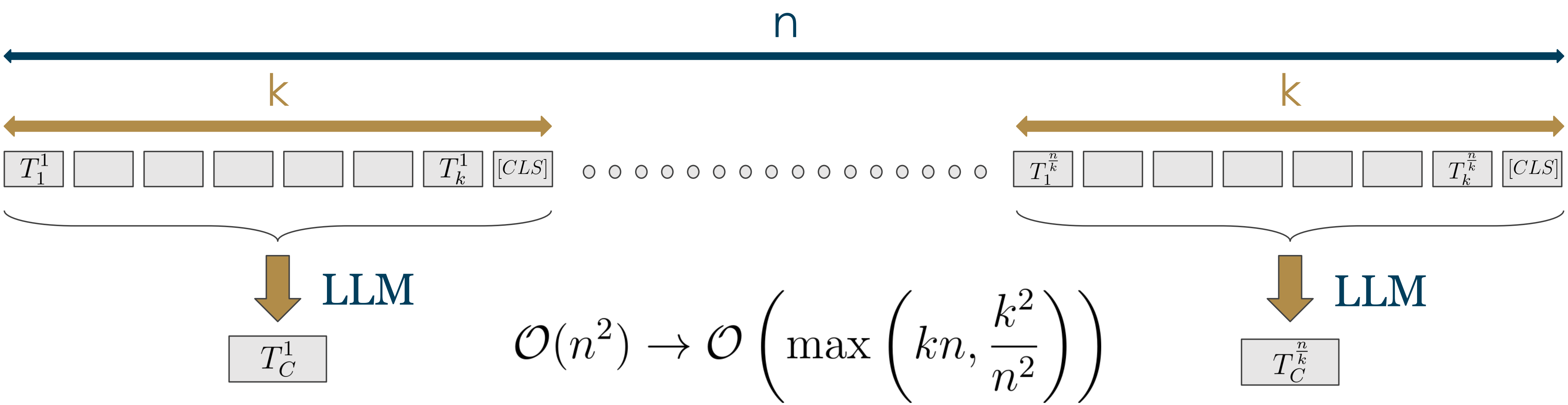
1 - Background

Hierarchical Transformers (Pappagari et al., 2019)

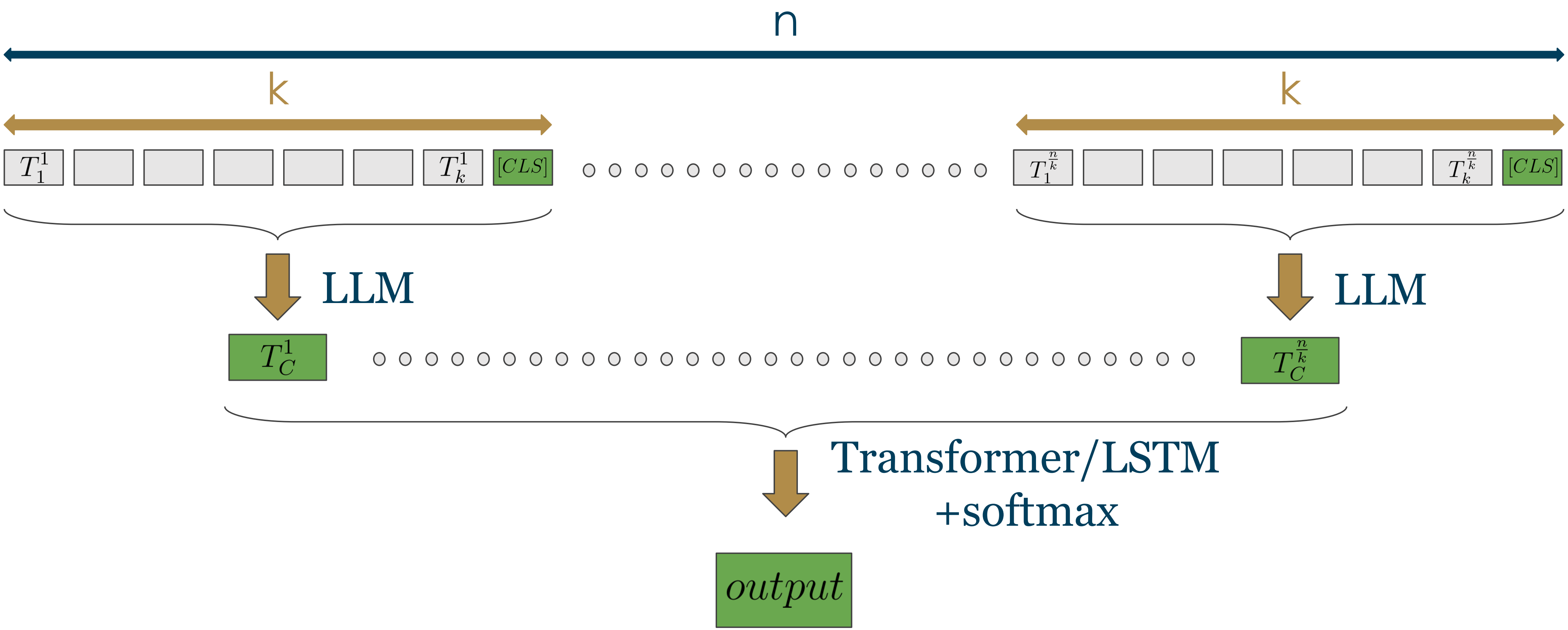
Text classification / Customer satisfaction prediction after a call

Improving the efficiency of text analysis



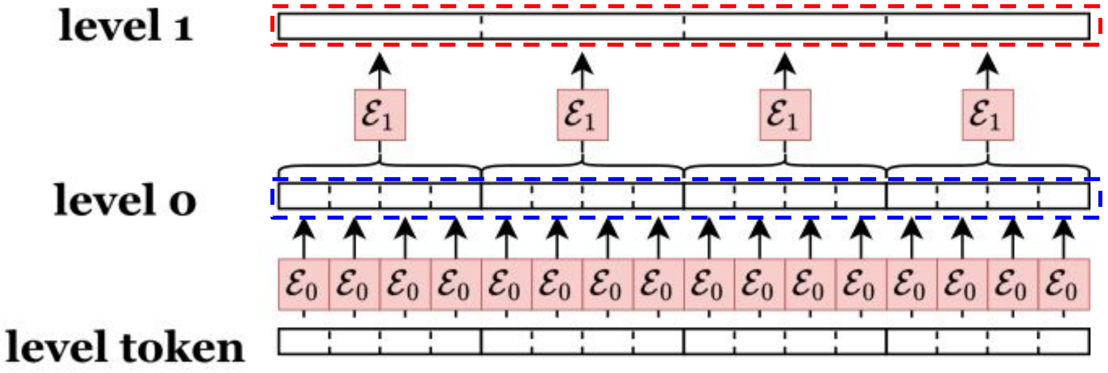
$$\mathcal{O}(n^2) \rightarrow \mathcal{O}\left(\max\left(kn, \frac{k^2}{n^2}\right)\right)$$

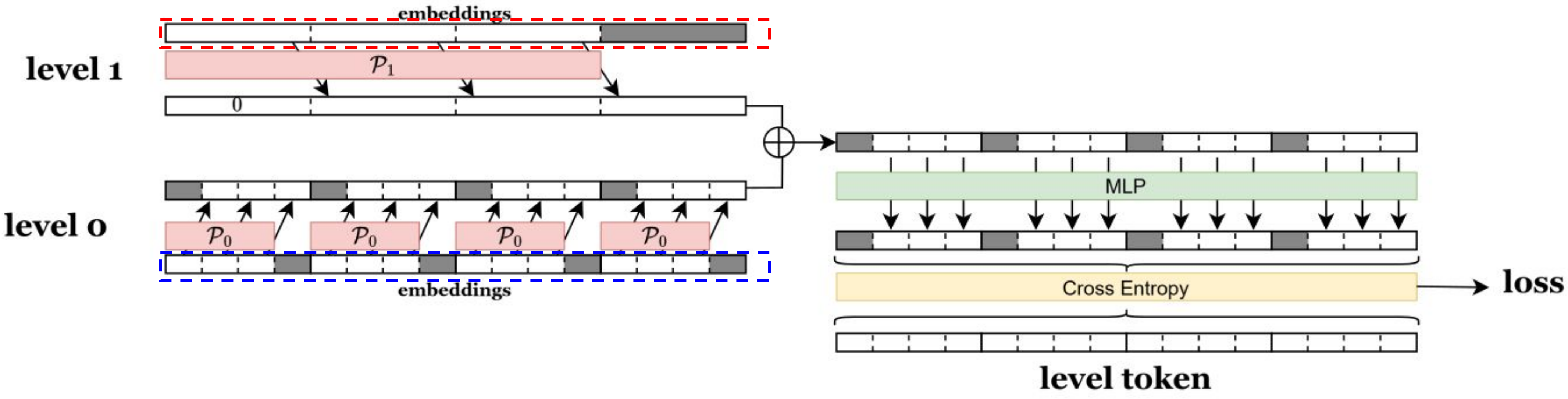
Hierarchical Transformers (Pappagari et al., 2019)



2 - Proposed architecture

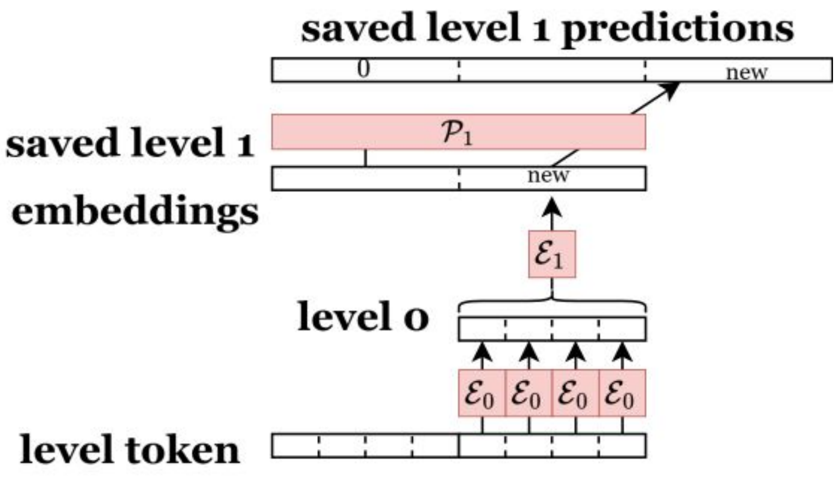
Training :



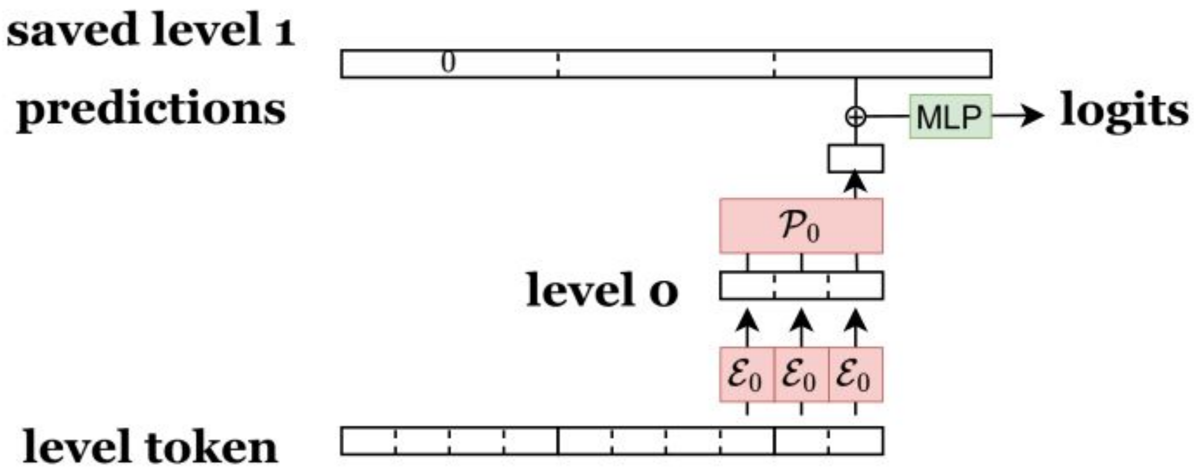


2 - Proposed architecture

Inference :



Level 1 update



Token prediction

2 - Proposed architecture

$\mathcal{E}$ **and** $\mathcal{P}$ → **based on GPT2**

initial weights are kept

$\mathcal{E}_0$ Token embedder of GPT2

$\mathcal{P}_0$ Transformers of GPT2

$\mathcal{E}_1$ Transformers of GPT2 with a CLS token

$\mathcal{P}_1$ Transformers of GPT2

MLP = Last linear layer final of GPT2

→ All of these networks are finetuned

3 - Results and possible improvements

Training: batch_size=32 ; context_size_0 = 16 ; context_size_1 = 8

Validation: Loss identical to the training loss on a validation set (20 batches)

Method	Loss
No hierarchy	3.80
Inside sum	3.76
Outside sum	3.73
Outside sum and shared E1/P0	3.73

Results

Dataset used: state of the union

3 - Results and possible improvements

- **Embedding aggregation**: Concatenation or summation?
- **Result weighting**: Should the outputs of the different embeddings be weighted?
- **Respective embedding dimensions**: Should different dimensions be used for the latent spaces?
- **Loss**: Should an intermediate loss be added on the latent space?
- **Overlapping**: Should training use overlapping windows ?

CONCLUSION

- **Idea:** Separate prediction computations across multiple hierarchical levels → improved complexity and abstraction
- **Results:** Encouraging, but at a small scale → what would the outcome be on larger datasets? What about the quality of the embeddings?
- **Future directions:** Numerous possibilities for improvement and adaptation

Bibliography

Pappagari, Raghavendra et al. (2019). *Hierarchical Transformers for Long Document Classification*. arXiv: 1910.10781 [cs.CL]. URL: https://arxiv.org/abs/1910.10781.

Yang, Zhilin et al. (2020). *XLNet: Generalized Autoregressive Pretraining for Language Understanding*. arXiv: 1906.08237 [cs.CL]. URL: https://arxiv.org/abs/1906.08237.

THANK YOU